

Boyi Li

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Education

Zhejiang University

Sep. 2023 – May. 2027

Bachelor of Science in Computer Engineering, Dual Degree with UIUC

Haining, Zhejiang

- **Relevant Coursework:** Data Structures (CS 225), Analog Signal Processing (ECE 210), Computer Systems & Programming (ECE 220), Linear Algebra with Computational Applications (MATH 257), Basic Discrete Mathematics (MATH 213), Probability with Eng App (ECE 313), etc.

University of Illinois at Urbana-Champaign (UIUC)

Sep. 2023 – May. 2027

Bachelor of Science in Computer Engineering, Dual Degree with ZJU

Champaign, Illinois

- **Relevant Coursework:** Machine Learning (CS 446), Applied Machine Learning (CS 441), Artificial Intelligence (CS 440), Computer Systems Engineering (ECE 391), Digital Systems Laboratory (ECE 385), Intro Differential Equations (MATH 285), etc.

Publications

Zhonghan Zhao, Wenhao Chai, Xuan Wang, **Boyi Li**, Shengyu Hao, Shidong Cao, Tian Ye, Gaoang Wang. **See and Think: Embodied Agents in Virtual Environments**. In *European Conference on Computer Vision (ECCV)*. 2024.

Boyi Li, Zhonghan Zhao, Der-Horng Lee, Gaoang Wang. **Adaptive Graph Pruning for Multi-Agent Communication**. In *European Conference on Artificial Intelligence (ECAI)*. 2025. **Spotlight Paper**.

Boyi Li, Yifan Shen, Houze Yang, Xu Cao, Guojun Yun, Li Gao, Turong Chen, Long Xu, Jianguo Cao, Meihuan Huang. **The 1st AI Children Challenge**. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*. 2026.

Research Experience

Rehg Lab, UIUC

Sep. 2025 – Present

Research Intern

Champaign, Illinois

Advisor: *Prof. James M. Rehg*

Conducted research on Multimodal LLMs.

- Investigated multimodal LLMs for improving abstract reasoning in vision-language settings.
- Proposed a training framework named Cognitive Supersensing, incorporating structured visual representations, achieving up to 33.5% improvement over strong baselines.

Conducted research on Computer Vision for Healthcare.

- Studied fine-grained pediatric gait analysis from video for clinically relevant motion understanding.
- Curated a pediatric gait dataset and developed a model achieving state-of-the-art performance on benchmark tasks.

CVNext Lab, Zhejiang University

Sep. 2023 – Aug 2025

Research Intern

Haining, Zhejiang

Advisor: *Prof. Gaoang Wang*

Conducted research on Embodied AI.

- Proposed STEVE, a comprehensive and visionary embodied agent in the Minecraft virtual environment, with a corresponding dataset that includes vision-environment, knowledge question-answering, and skill-code pairs.
- STEVE achieves up to 1.5x faster unlocking key tech trees and 2.5x quicker in block search tasks.

Conducted research on Multi-Agent Systems.

- Proposed Adaptive Graph Pruning (AGP), a novel task-adaptive multi-agent collaboration framework that jointly optimizes agent quantity (hard-pruning) and communication topology (soft-pruning).
- AGP achieves an increase in performance of 2.58% - 9.84% across six mainstream benchmarks with a decrease in token consumption of 90%+.

- Investigated sketch-based interaction methods for lowering the barrier to 3D design in virtual reality environments.
- Proposed a system combining freehand VR input with model recognition to enable automatic reconstruction of structured 3D geometry.
- Conducted preliminary user evaluations showing improved efficiency and usability for novice users.

Service

Workshop Co-Organizer

CVPR 2026

The 2nd CVPR 2026 Workshop on Computer Vision for Children

- Co-organizing the 2nd Computer Vision for Children Workshop, responsible for overall organization and coordination.
- Leading the organization the 1st AI for Children Challenge, including dataset curation and evaluation design.

Projects

Unix-like Operating System

- **Role:** Project lead. Primarily responsible for the UART, VirtIO, and File System components.
- Developed a Unix-like operating system supporting user program execution, virtual memory management, and system calls, using C, RISC-V assembly, and Sv39 paging.
- Implemented virtual memory subsystem with multi-level page tables, demand paging, and secure memory isolation for user processes.
- Implemented process abstraction and ELF loader, enabling dynamic program loading, trap-based user/kernel transitions.

BioElectro Glove: Mental Health Screening through Connective Tissue Signals

- **Role:** Tech Lead. Responsible for product architecture design and model training.
- Based on fascia theory, this project explores depression diagnosis via hand bioelectrical signals, leveraging the physiological link between emotional states, connective tissue health, and oxidative stress, enabling a non-invasive and repeatable detection method.
- The project won a Gold Award in the China International College Students' Innovation Competition.

Honors and Awards

Gold Award

2025

China International College Students' Innovation Competition (CICSIC) 2025

Gold Award

2025

Singapore Division Contest of China International College Students' Innovation Competition (CICSIC-SG) 2025

Honorable Mention

2024

Mathematical Contest In Modeling (MCM)

Third-Class Academic Excellence Scholarship

2024 – 2025

Zhejiang University

Student Outstanding Academic Achievement Award

2024 – 2025

Zhejiang University

Student Innovation and Entrepreneurship Award

2023 – 2024, 2024 – 2025

Zhejiang University

Languages and Skills

Chinese: Native

English: Proficient (TOEFL 5/6 (100/120))

Programming Languages: C/C++, Python, C#, JavaScript

Hardware & Embedded Systems: FPGA, Intel Quartus